

Linear Models Lecture 6: Heteroskedasticity

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Mean Independence Does Not Rule Out Heteroskedasticity

Let

$$X \sim \text{Uniform}(-1, 1), \quad Y = X^2 + X\varepsilon,$$

where $\varepsilon \sim \mathcal{N}(0, 1)$ and $\varepsilon \perp X$.

The CEF is $m(X) = X^2$, and the CEF error is $e = X\varepsilon$.

Mean independence holds: $\mathbb{E}[e|X] = X\mathbb{E}[\varepsilon|X] = 0$.

But the conditional variance depends on X :

$$\text{Var}(e|X) = X^2.$$

$\mathbb{E}[e|X] = 0$ guarantees that e has no *systematic* pattern with X . It says nothing about the *spread* of e . Heteroskedasticity is the norm, not the exception.

Theory-Based vs. Agnostic Strategies

- Two approaches to handling heteroskedasticity:
 - 1 **Theory-based (GLS):** Model the variance structure Σ and use it to improve efficiency.
 - Requires knowing (or estimating) *why* some observations are noisier.
 - Example: if the mechanism is proportional growth, take logs—this may eliminate heteroskedasticity at the source (recall the invariance discussion from Lecture 5).
 - If correct, GLS is more efficient than OLS with robust SEs.
 - 2 **Agnostic (robust SEs):** Use OLS but correct the standard errors for unknown heteroskedasticity.
 - Requires no model for Σ —valid under any form of heteroskedasticity.
 - Sacrifices efficiency for robustness.
- In practice, most applied work uses robust SEs as the default and reserves GLS for settings where theory strongly constrains Σ .

Why homoskedastic SEs can be dangerously wrong (Hansen 4.13)

Under homoskedasticity, we estimate $\hat{\mathbf{V}}_{\hat{\beta}}^0 = (\mathbf{X}'\mathbf{X})^{-1}s^2$. But the **true** variance under heteroskedasticity is $\mathbf{V}_{\hat{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\mathbf{D}\mathbf{X})(\mathbf{X}'\mathbf{X})^{-1}$.

Example (Hansen): Suppose $k = 1$, $\sigma_i^2 = X_i^2$, and $\mathbb{E}[X] = 0$. Then:

$$\frac{V_{\hat{\beta}}}{\mathbb{E}[\hat{\mathbf{V}}_{\hat{\beta}}^0]} \approx \frac{\mathbb{E}[X^4]}{(\mathbb{E}[X^2])^2} \stackrel{\text{def}}{=} \kappa$$

- If $X \sim N(0, \sigma^2)$: $\kappa = 3$. True variance is **3×** the homoskedastic estimate.
- For *wage* in the CPS: $\kappa = 30$. True variance is **30×** the homoskedastic estimate.
- The homoskedastic SE understates uncertainty by a factor of $\sqrt{\kappa}$.

Takeaway: The classical covariance estimator can be wildly misleading. Always use a heteroskedasticity-robust estimator.

“Agnostic” Covariance Matrix Estimation under heteroskedasticity

Suppose we have heteroskedasticity:

$$\begin{aligned} \text{var}(\hat{\beta}) &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{D}\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} \\ \mathbf{D} &= \text{diag}(\sigma_1^2, \dots, \sigma_n^2) \end{aligned}$$

If we knew $e_1^2, \dots, e_n^2$, then we could just plug them in:

$$\tilde{\text{var}}(\hat{\beta}) = (\mathbf{X}'\mathbf{X})^{-1} \left(\sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i' e_i^2 \right) (\mathbf{X}'\mathbf{X})^{-1}$$

If we use observed $\hat{e}^2$, then this is called the HC0 (heteroskedasticity consistent) estimator, or White covariance matrix estimator.

From the Sandwich to HC Estimators

Recall from Lecture 5: the true variance under heteroskedasticity is the **sandwich**:

$$\mathbf{V}_{\hat{\beta}} = (\mathbf{X}'\mathbf{X})^{-1} \underbrace{\left(\sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i' \sigma_i^2 \right)}_{\text{"meat" } \mathbf{X}'\mathbf{D}\mathbf{X}} (\mathbf{X}'\mathbf{X})^{-1}$$

We never observe $\sigma_i^2 = \text{Var}(e_i|\mathbf{X})$. Each HC variant estimates it differently:

Estimator	Plug-in for σ_i^2	Rationale
HC0	$\hat{e}_i^2$	Simplest; biased downward
HC1	$\frac{n}{n-k} \hat{e}_i^2$	Degrees-of-freedom correction
HC2	$\frac{\hat{e}_i^2}{1-h_{ii}}$	Unbiased: $\mathbb{E}[\hat{e}_i^2 \mathbf{X}] = (1-h_{ii})\sigma_i^2$
HC3	$\frac{\hat{e}_i^2}{(1-h_{ii})^2}$	Conservative (jackknife-like)

Problems with White covariance estimators

- HC0 is downward biased in finite samples.
 - It is missing the $\frac{1}{n-k}$ term for degrees of freedom.
- Fixing this gets us HC1.
- It fails to account for leverage:

$$\mathbb{E}[\hat{e}_i^2] = \sigma_i^2(1 - h_{ii}) < \sigma_i^2$$

- Fixing this gets us HC2

HC2 and HC3: Leverage-corrected estimators

HC2 (unbiased) and HC3 (conservative) correct for the leverage of each observation:

$$\hat{\mathbf{V}}_{\hat{\beta}}^{HC2} = (\mathbf{X}'\mathbf{X})^{-1} \left(\sum_{i=1}^n (1 - h_{ii})^{-1} \mathbf{x}_i \mathbf{x}_i' \hat{e}_i^2 \right) (\mathbf{X}'\mathbf{X})^{-1}$$

$$\hat{\mathbf{V}}_{\hat{\beta}}^{HC3} = (\mathbf{X}'\mathbf{X})^{-1} \left(\sum_{i=1}^n (1 - h_{ii})^{-2} \mathbf{x}_i \mathbf{x}_i' \hat{e}_i^2 \right) (\mathbf{X}'\mathbf{X})^{-1}$$

Since $(1 - h_{ii})^{-2} > (1 - h_{ii})^{-1} > 1$, we always have:

$$\hat{\mathbf{V}}_{\hat{\beta}}^{HC0} < \hat{\mathbf{V}}_{\hat{\beta}}^{HC2} < \hat{\mathbf{V}}_{\hat{\beta}}^{HC3}$$

Implementation in R

Option 1: estimatr package (recommended for applied work)

```
library(estimatr)
# HC2 is the default -- unbiased under homoskedasticity
lm_robust(y ~ x1 + x2, data = dta, se_type = "HC2")
# HC3 is conservative (biased away from zero)
lm_robust(y ~ x1 + x2, data = dta, se_type = "HC3")
```

Option 2: sandwich + lmtest (flexible, works with any lm object)

```
library(sandwich); library(lmtest)
model <- lm(y ~ x1 + x2, data = dta)
coeftest(model, vcov = vcovHC(model, type = "HC2"))
```

R Example: Comparing Standard Errors (Hansen 4.15)

```
library(estimatr); library(sandwich); library(lmtest)
data(mtcars)
m <- lm(mpg ~ wt + hp, data = mtcars)

# Homoskedastic (classical) SEs
se_classical <- summary(m)$coefficients[, "Std. Error"]

# Robust SEs using sandwich
se_hc0 <- sqrt(diag(vcovHC(m, type = "HC0")))
se_hc1 <- sqrt(diag(vcovHC(m, type = "HC1")))
se_hc2 <- sqrt(diag(vcovHC(m, type = "HC2")))
se_hc3 <- sqrt(diag(vcovHC(m, type = "HC3")))

cbind(Classical = se_classical, HC0 = se_hc0,
      HC1 = se_hc1, HC2 = se_hc2, HC3 = se_hc3)
```

Note: $HC0 < HC2 < HC3$ always holds. The differences grow when observations have high leverage (h_{ii} close to 1).

Simulation: Classical SEs Undercover

```
set.seed(1); B <- 5000; n <- 100; cover_cl <- cover_hc2 <- 0
for (b in 1:B) {
  x <- rexp(n, 1); e <- rnorm(n, 0, sd = x) # sigma_i = x_i
  y <- 1 + 2*x + e; m <- lm(y ~ x)
  ci_cl <- confint(m)["x",] # classical
  ci_hc2 <- coefci(m, vcov.=vcovHC(m, "HC2"))["x",] # robust
  cover_cl <- cover_cl + (2 > ci_cl[1] & 2 < ci_cl[2])
  cover_hc2 <- cover_hc2 + (2 > ci_hc2[1] & 2 < ci_hc2[2])
}
c(Classical = cover_cl/B, HC2 = cover_hc2/B)
# Classical ~ 0.83 HC2 ~ 0.95
```

Classical 95% CIs cover the true β only $\sim 83\%$ of the time. HC2 CIs achieve the nominal 95% coverage.

Notes on Interpretation

- Robust SEs do **not** affect coefficient estimates or R^2 /RMSE.
- They only change standard errors, t -statistics, and confidence intervals.
- Use Wald tests (not classical F -tests) for joint hypotheses:

```
library(car)
linearHypothesis(model, c("x1 = 0", "x2 = 0"),
                  white.adjust = "hc2")
```

- Hansen recommends HC2 (unbiased under homoskedasticity) or HC3 (conservative for any $\mathbf{X}$) over HC1.
- In most applications HC1, HC2, HC3 are similar. They diverge when some h_{ii} is large (high-leverage observations).

HC1 does not work with Sparse Dummy Variables

- Suppose $Y = \beta_1 D + \beta_2 + e$, where $D_i = 1$ for n_1 cases.
- In the extreme case, $n_1 = 1$:

$$V_{\hat{\beta}} = \sigma^2 (\mathbf{X}'\mathbf{X})^{-1} = \sigma^2 \begin{pmatrix} 1 & 1 \\ 1 & n \end{pmatrix}^{-1} = \sigma^2 \frac{1}{n-1} \begin{pmatrix} n & -1 \\ -1 & 1 \end{pmatrix}$$

$$V_{\hat{\beta}_1} = \sigma^2 \frac{n}{n-1}$$

- Consider the estimator $\hat{\theta} = \hat{\beta}_1 + \hat{\beta}_2$, with variance $\sigma^2 \frac{n}{n-1} + \sigma^2 \frac{1}{n-1} - \sigma^2 \frac{2}{n-1} = \sigma^2$

$$\hat{V}_{\beta}^{HC1} = s^2 \frac{n}{(n-1)^2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$$

$$\hat{V}_{\theta}^{HC1} = \underbrace{s^2 \frac{n}{(n-1)^2}}_{V_{11}} - \underbrace{s^2 \frac{n}{(n-1)^2}}_{V_{12}} - \underbrace{s^2 \frac{n}{(n-1)^2}}_{V_{21}} + \underbrace{s^2 \frac{n}{(n-1)^2}}_{V_{22}} = 0$$

Measures of Fit (Hansen 4.18)

- $R^2 = 1 - \frac{\hat{\sigma}^2}{\hat{\sigma}_Y^2}$ *always increases* when regressors are added $\rightarrow$ cannot be used for model selection.
- $\bar{R}^2$ (adjusted) corrects with $(n-1)/(n-k)$, but Hansen argues it still tends to select models with too many parameters.
- **Recommended:** the leave-one-out cross-validation $\tilde{R}^2$:

$$\tilde{R}^2 = 1 - \frac{\sum_{i=1}^n \tilde{e}_i^2}{\sum_{i=1}^n (Y_i - \bar{Y})^2} = 1 - \frac{\bar{\sigma}^2}{\hat{\sigma}_Y^2}$$

where $\tilde{e}_i = \hat{e}_i / (1 - h_{ii})$ are the prediction errors.

- $\tilde{R}^2$ estimates the percentage of forecast variance explained; it can be *negative* if the model predicts worse than the mean.
- Hansen: “It is recommended to omit R^2 and $\bar{R}^2$. If a measure of fit is desired, report $\tilde{R}^2$ or $\bar{\sigma}^2$.”

Clustered Standard Errors

- Suppose that our data are drawn from groups.
E.g. students in a school, districts in a state, apartments in a building.
- These observations are subject to common shocks (even if observations don't affect one another).
- These data are called *clustered*.
- We will assume that clusters are known to the researcher and observations are independent across clusters.

Motivating Example: Duflo, Dupas, and Kremer (2011)

In 2005, 140 primary schools in Kenya received funding to hire an extra teacher. Half assigned students to classrooms by prior test score (“tracking”).

$$\widehat{TestScore}_{ig} = -0.071 + 0.138 \text{ Tracking}_g + \hat{e}_{ig}$$

- With **conventional robust SEs**: $s(\hat{\gamma}) = 0.026$
- With **cluster-robust SEs** (at school level): $s(\hat{\gamma}) = 0.078$

The cluster-robust SEs are **3× larger** than the conventional ones!

Ignoring clustering would vastly overstate the precision of the estimated treatment effect. The cluster-robust standard error is the appropriate one because student achievement within a school is correlated.

Formalism for Clustered Regression

- $(Y_{ig}, \mathbf{x}_{ig})$ where $g = 1, \dots, G$ indexes a cluster and $i = 1, \dots, n_g$ indexes individuals in cluster g .
- $\mathbf{Y}_g = (Y_{1g}, \dots, Y_{n_g, g})'$, $\mathbf{X}_g = (\mathbf{x}_{1g}, \dots, \mathbf{x}_{n_g, g})'$ at the group level.

$$Y_{ig} = \mathbf{x}_{ig}'\beta + e_{ig}$$

$$\begin{aligned}\hat{\beta} &= \left(\sum_{g=1}^G \sum_{i=1}^{n_g} \mathbf{x}_{ig} \mathbf{x}_{ig}' \right)^{-1} \left(\sum_{g=1}^G \sum_{i=1}^{n_g} \mathbf{x}_{ig} Y_{ig} \right) \\ &= \left(\sum_{g=1}^G \mathbf{X}_g' \mathbf{X}_g \right)^{-1} \left(\sum_{g=1}^G \mathbf{X}_g' \mathbf{y}_g \right) \\ &= (\mathbf{X}'\mathbf{X})^{-1} (\mathbf{X}'\mathbf{y})\end{aligned}$$

Variance of Clustered Regression

Call $\Sigma_g = E[\mathbf{e}_g \mathbf{e}_g']$ the $n_g \times n_g$ covariance in the g cluster.

$$\begin{aligned} \text{var} \left[\left(\sum_{g=1}^G \mathbf{X}_g' \mathbf{e}_g \right) \right] &= \sum_{g=1}^G \text{var} [\mathbf{X}_g' \mathbf{e}_g] && \text{(By independence across cluster)} \\ &= \sum_{g=1}^G \mathbf{X}_g' \text{var} [\mathbf{e}_g] \mathbf{X}_g \\ &= \sum_{g=1}^G \mathbf{X}_g' \Sigma_g \mathbf{X}_g \\ &\equiv \Omega_n \\ \mathbf{V}_{\hat{\beta}} &= (\mathbf{X}' \mathbf{X})^{-1} \Omega_n (\mathbf{X}' \mathbf{X})^{-1} \end{aligned}$$

Moulton (1990) Formula

Suppose all clusters are equal size N , homoskedastic within cluster $\mathbb{E}[e_{ig}^2] = \sigma^2$, with intra-cluster correlation $\mathbb{E}[e_{ig}e_{\ell g}] = \sigma^2\rho$ for $i \neq \ell$, and X_{ig} does not vary within a cluster:

$$\mathbf{V}_{\hat{\beta}} = (\mathbf{X}'\mathbf{X})^{-1} \sigma^2(1 + \rho(N - 1))$$

The **inflation factor** $1 + \rho(N - 1)$ can be enormous:

ρ	Cluster size N	Inflation factor
0.05	20	1.95
0.10	48	5.7
0.25	48	12.75
0.25	100	25.75

Even modest intra-cluster correlation with moderate cluster sizes produces large distortions in conventional SEs.

Strategy 1: Modeling intra-cluster dependence

- A random-effects model assumes:

$$u_{gi} = \lambda e_g + e_{gi}$$

- Where $e_{gi} \sim f(0, \omega^2)$ is individual specific, $e_g \sim f(0, 1)$ is cluster wide and the two are independent.

$$\Omega_g = \begin{bmatrix} \lambda^2 + \omega^2 & \lambda^2 & \dots & \lambda^2 \\ \lambda^2 & \lambda^2 + \omega^2 & \dots & \lambda^2 \\ \lambda^2 & \lambda^2 & \dots & \lambda^2 + \omega^2 \end{bmatrix}$$

FGLS

- Estimate pooled model of $\mathbf{y}$ on $\mathbf{X}$ using OLS. Record $\hat{\sigma}^2$
- Estimate model with fixed effects: regress $\mathbf{y}$ on $[\mathbf{X} \ \mathbf{D}]$. This is our estimate of ω^2 , $\hat{\omega}^2$.
- $\hat{\lambda}^2 = \hat{\sigma}^2 - \hat{\omega}^2$.
- Plug into GLS for Σ .
- The Random Effects Estimator is consistent, biased, and asymptotically efficient (if the errors are correct).
- However, if we needed group fixed effects in $\mathbf{X}$, we already estimate λe_g , and this model will not work.

Intra-cluster dependence that survives fixed effects

- Contrast the random effects model with a factor model: $u_{gi} = \lambda_{gi}e_g + e_{gi}$, where now λ_{gi} depends on the individual (e.g., some students are affected more by teacher quality than others).
- If we use cluster fixed effects:

$$u_{gi} - \bar{u}_g = (\lambda_{gi} - \bar{\lambda}_g)e_g + e_{gi} - \bar{e}_g$$

$$\text{cov}(u_{gi} - \bar{u}_g, u_{gl} - \bar{u}_g) = (\lambda_{gi} - \bar{\lambda}_g)(\lambda_{gl} - \bar{\lambda}_g)$$

- This is zero only if λ_{gi} is the same for all i — fixed effects remove the common shock but not heterogeneous exposure to it.
- Unfortunately, it is not obvious how to implement this model.

Strategy 2: Cluster Robust Standard Errors Arellano (1987), Hansen (2007)

The squared error e_i^2 is an unbiased estimate for $E[e_i^2]$, $\mathbf{e}_g \mathbf{e}_g'$ is unbiased for $E[\mathbf{e}_g \mathbf{e}_g']$. As with White, we can estimate e_i^2 with $\hat{e}_i^2$:

$$\begin{aligned}\hat{\Omega}_n &= \sum_{g=1}^G \mathbf{x}_g' \hat{\mathbf{e}}_g \hat{\mathbf{e}}_g' \mathbf{x}_g \\ \hat{\mathbf{V}}_{\hat{\beta}} &= a_n (\mathbf{X}' \mathbf{X})^{-1} \hat{\Omega}_n (\mathbf{X}' \mathbf{X})^{-1} \\ a_n &= \left(\frac{n-1}{n-k} \right) \left(\frac{G}{G-1} \right)\end{aligned}$$

a_n is the small sample correction used by STATA, recommended by Hansen (2007).

Clustered SEs in R

Option 1: estimatr (simplest)

```
library(estimatr)
lm_robust(y ~ x1 + x2, data=dta, clusters=cluster_id, se_type="CR2")
```

Option 2: sandwich + lmtest

```
library(sandwich); library(lmtest)
model <- lm(y ~ x1 + x2, data = dta)
coeftest(model, vcov = vcovCL(model, cluster = dta$cluster_id))
```

Option 3: Manual construction (Hansen Ch. 4 R code)

```
xe <- x * rep(e, times = k)           # X_i * e_i
xe_sum <- rowsum(xe, cluster_id)      # sum within clusters
G <- nrow(xe_sum); omega <- t(xe_sum) %*% xe_sum
scale <- G/(G-1) * (n-1)/(n-k)
V_cl <- scale * invx %*% omega %*% invx
```


Inference with Clustered Samples (Hansen 4.22)

- The **effective sample size** for cluster-robust inference is G (number of clusters), **not** n (number of observations).
- The cluster-robust estimator treats each cluster as a single observation and estimates the covariance from the variation across cluster means.
- If $G = 50$, inference quality is comparable to heteroskedasticity-robust inference with $n = 50$.
- Most cluster-robust theory assumes **homogeneous** cluster sizes. When cluster sizes are highly unequal, cluster sums have heterogeneous variances $\rightarrow$ compounding problem.
- When the number of *treated* clusters is small (e.g. only a few schools received treatment), the cluster-robust SE on the treatment coefficient can be severely downward biased—analogueous to the sparse dummy variable problem (Section 4.16).

At what level ought one cluster? (Hansen 4.23)

- Should we cluster by individual, county, state, or region?
- There is a **bias–variance tradeoff**:
 - **Too fine** (e.g. household instead of village): omits covariance terms → SEs biased *downward*, spurious significance.
 - **Too coarse** (e.g. state instead of county): adds noise → SEs imprecise, less power.
- Rules of thumb:
 - Cluster at the level where treatment is assigned.
 - Cluster at the coarsest level defensible by theory, provided G is not too small ($G \geq 50$ preferred).
 - Your effective sample size is G , not n .
- Honest assessment (Hansen): “We really do not know what is the ‘correct’ level at which to do cluster-robust inference.”

Practical Recommendations (Hansen Ch. 4 Summary)

- 1 Always report robust standard errors.** The classical homoskedastic formula $s^2(\mathbf{X}'\mathbf{X})^{-1}$ is only valid under the strong (and rarely true) assumption of conditional homoskedasticity.
- 2 Use HC2 by default** for cross-sectional data. It is unbiased under homoskedasticity and performs well under heteroskedasticity. Use HC3 if you want a conservative alternative.
- 3 If data are clustered**, use cluster-robust SEs. Your effective sample size is G (clusters), not n (observations). Cluster at the coarsest defensible level.
- 4 For measures of fit**, prefer $\tilde{R}^2$ (leave-one-out cross-validation) over R^2 or $\bar{R}^2$. Hansen: "It is recommended to omit R^2 and $\bar{R}^2$."
- 5 Watch for sparse dummies and high leverage.** These can cause robust SE estimators to break down or be severely biased.

Multicollinearity (Hansen 4.20)

- **Strict multicollinearity:** $\mathbf{X}'\mathbf{X}$ is singular $\rightarrow \hat{\beta}$ is undefined. Typically caused by redundant fixed effects or a dummy variable trap.
- **Near multicollinearity:** regressors are highly correlated. With two regressors and correlation ρ :

$$\text{var}[\hat{\beta}_j|\mathbf{X}] = \frac{\sigma^2}{n(1 - \rho^2)}$$

As $\rho \rightarrow 1$, the variance $\rightarrow \infty$.

- This is *not* a bias problem—it is purely a precision problem, equivalent to having a small sample (Goldberger's “micronumerosity” critique).
- Robust standard errors can be **sensitive to high leverage** under near multicollinearity, producing misleadingly small SEs even when coefficient estimates are imprecise.

Ridge Regression: Trading Bias for Variance

- Near-multicollinearity inflates $\text{Var}(\hat{\beta})$ because $(\mathbf{X}'\mathbf{X})^{-1}$ has large entries. Ridge adds a penalty that shrinks coefficients toward zero:

$$\hat{\beta}_{\text{ridge}} = (\mathbf{X}'\mathbf{X} + \lambda \mathbf{I}_k)^{-1} \mathbf{X}'\mathbf{y}$$

- This is **biased**: $\mathbb{E}[\hat{\beta}_{\text{ridge}}|\mathbf{X}] = (\mathbf{X}'\mathbf{X} + \lambda \mathbf{I}_k)^{-1} \mathbf{X}'\mathbf{X}\beta \neq \beta$.
- But the variance is smaller: adding $\lambda \mathbf{I}_k$ ensures $(\mathbf{X}'\mathbf{X} + \lambda \mathbf{I}_k)$ is always well-conditioned, even if $\mathbf{X}'\mathbf{X}$ is nearly singular.
- The tradeoff: $\lambda = 0$ gives OLS (unbiased, high variance); large λ gives heavy shrinkage (biased, low variance). Choose λ by cross-validation.
- In R: `glmnet::glmnet(X, y, alpha=0)` or `MASS::lm.ridge`.